1. Clinical Question & Core Hypothesis

Clinical Question

Why do tumors with abundant lymphocyte infiltration (high TILs) sometimes fail to show the expected overall survival benefit? The KG suggests intersections between tumor-intrinsic stress/ death programs (TP53/autophagy), immune-checkpoint regulation (LAG3), and clinical outcomes.
2) Tier 1 paths used: (a) TP53 is linked to urothelial carcinoma / tumor stress programs (TP53 ↔ cell-death/autophagy), and (b) BLCA context shows a protective/modulatory role for LAG3 (interpreted here as LAG3 being a key immune-checkpoint axis in tumor immune regulation). I built on these because they form a mechanistic backbone connecting tumor intrinsic state (TP53) to T-cell functional state (LAG3) that plausibly modulates the effect of TILs.
3) Tier 2 evidence supporting claim: bidirectional links between cell death and autophagy (cell-death ↔ autophagy) provide a mechanism by which TP53 dysfunction could reprogram tumor stress/metabolism and thereby alter immune signaling; KRAS influences cell-death but was not linked to OS (supporting deprioritization).
4) Why OS as endpoint: OS is the most robust, well-annotated endpoint in TCGA and in the KG (multiple edges reference OS and hazard ratios), so it best captures the clinical consequence of immune–tumor interactions.
5) Deliberately excluded elements and why: I excluded the infection/diabetes → perinephric abscess → hematogenous dissemination chain and BLCA-specific site implications because they are epidemiologic/organ-specific and would dilute a focused mechanistic hypothesis translatable to TCGA-BRCA; KRAS was relegated to secondary modifiers because it lacked a clear OS association in the KG and to keep the core hypothesis sharp (max 2 anchor genes)?

Working Hypothesis

Among breast cancers, high TIL density is associated with better OS in TP53-wildtype tumors but not in TP53-mutant tumors, where TP53 dysfunction reprograms autophagy-related stress pathways to promote T cell exhaustion via LAG3. Consequently, TIL-high/TP53-mutant tumors with high LAG3 expression will have worse OS than TIL-high/TP53-wildtype tumors with low LAG3, identifying a subset that may benefit from LAG3-directed immunotherapy.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	TP53, LAG3
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TP53: TP53 is linked to urothelial carcinoma / tumor stress programs (TP53 ↔ cell-death/autophagy), and (b) BLCA context shows a protective/modulatory role for LAG3 (interpreted here as LAG3 being a key immune-checkpoint axis in tumor immune regulation).

LAG3: LAG3 (interpreted here as LAG3 being a key immune-checkpoint axis in tumor immune regulation).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	KRAS	Supporting modifier from knowledge graph
Secondary	autophagy	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- KRAS: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- autophagy: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.